

Hannah Brown

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Education

Loyola University Chicago

Chicago, IL

Bachelor of Science in Bioinformatics, Biostatistics Minor (3.57)

August 2022 – May 2025

Awards: Graduated Cum Laude, Dean's list Spring 2023 – Spring 2025

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computational Biology, Intro & Advanced Biostatistics, SAS Programming & Applied Statistics, Bioinformatics, Genomics, Exploring Proteins, Molecular Biology Lab

Skills

Python, JavaScript, R, SAS, SQL, pandas, matplotlib, data cleaning, pipeline development, workflow automation, statistical analysis, data visualization, Bowtie2, SPAdes, BLAST, debugging, documentation, reproducibility, technical communication, Linux/command line.

Coding Projects

scPrediXcan Gene Expression Prediction Pipeline • Python

Developed the automated wrapper script that orchestrated the full analysis pipeline, integrating and validating code written by multiple teammates. Managed data flow, execution order, and cross-module dependencies to ensure the pipeline ran reproducibly from raw inputs through final outputs. Demonstrates collaborative development, modular design, and research-grade pipeline engineering.

Transcriptome Processing Pipeline • Python

Built a fully automated, multi-stage workflow that downloads raw SRA data, converts and organizes FASTQ files, maps reads with Bowtie2, filters and converts alignments with Samtools, assembles transcripts with SPAdes, and performs BLAST analysis on assembled contigs. Implemented modular orchestration functions, custom logging, error handling, and reproducible directory and file management. Demonstrates end-to-end workflow automation, multi-tool integration, and robust pipeline engineering.

Research Experience

Student Researcher • Loyola University Chicago

August – December 2024

Designed and executed multi-step experimental workflows requiring precise documentation, troubleshooting, and reproducibility. Analyzed sequencing and PCR-based data to identify nine bacterial isolates, applying bioinformatic tools for sequence alignment, classification, and validation. Interpreted results, compared findings against reference databases, and communicated conclusions through written reports and presentations.

January – May 2024

Independently characterized an unknown clinical bacterial isolate using culture-based and nucleic acid-based analytical methods. Performed whole genome sequencing, analyzed sequencing data, and co-authored a peer-reviewed genome announcement (PMID: 39012132, PMCID: PMC11320926), demonstrating strong analytical reasoning and technical communication.

May – June 2023

Investigated DNA repair defects in mus109 Drosophila mutants by designing and evaluating controlled experiments involving oxidative stress exposure. Assessed experimental outcomes, synthesized findings with primary literature, and presented data-driven conclusions.

Additional Roles

Generative AI Strategist • Outlier AI

June 2024 – March 2025

Evaluated AI model limitations through strategic prompt engineering. Generated high-quality mathematical training data for next-generation language models. Skills: LaTeX, prompt engineering, model evaluation, technical writing.

Server

February 2022 – November 2024

Skills: Customer service, multitasking, conflict resolution, communication, time management, teamwork.